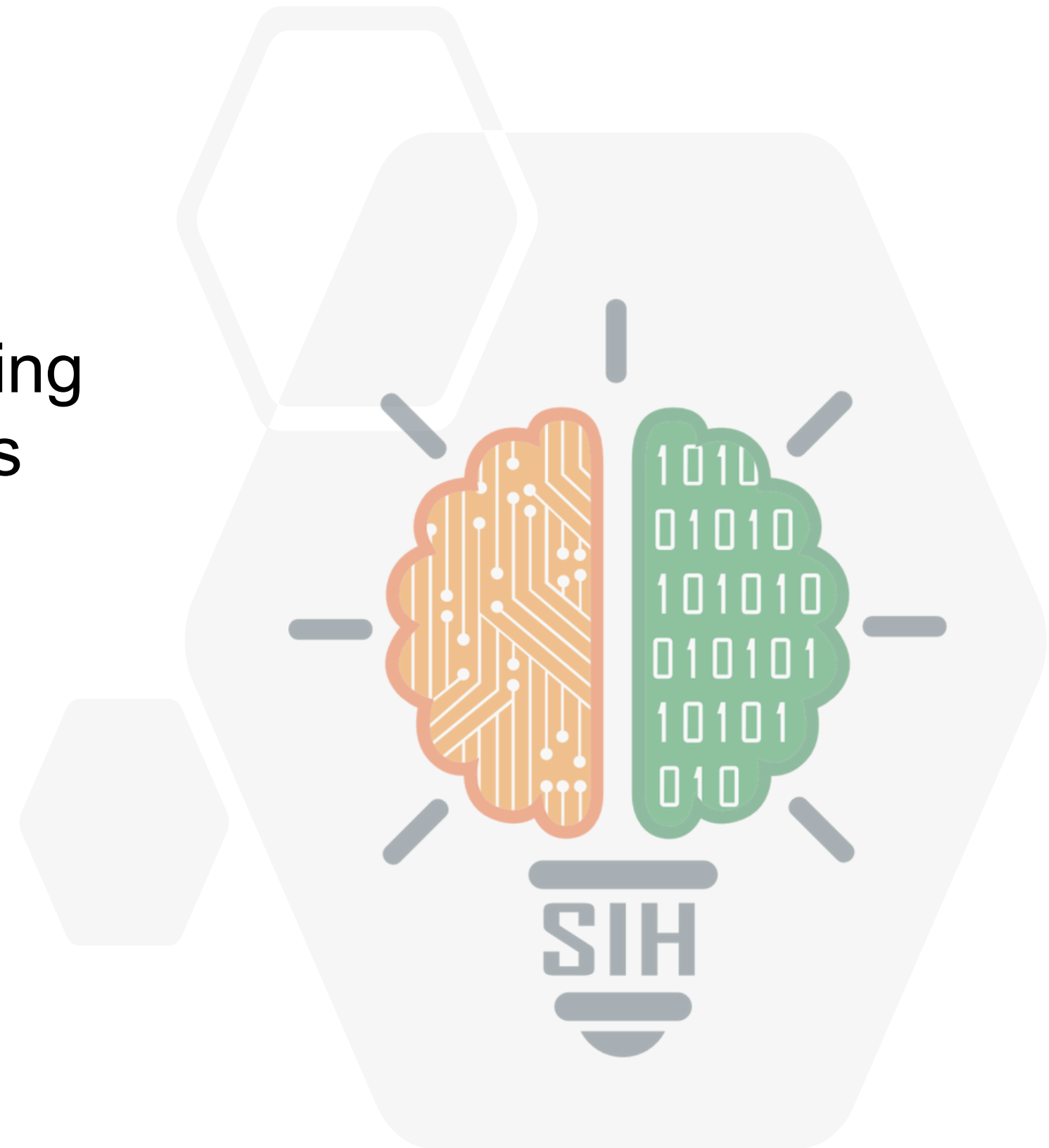


TITLE PAGE

- **Problem Statement ID** – SIH26037
- **Problem Statement Title** – Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- **Theme** – Robotics and Drones
- **PS Category** – Software
- **Team ID** – SIH26-A0H-T350
- **Team Name (Registered on portal)** – Pixels



Problem at Hand



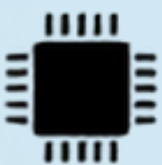
Unstructured Roads: Most Indian rural roads lack lane markings, have mixed traffic, uneven surfaces and frequent obstacles, making reliable navigation difficult.



Binary Occupancy Limitations: Traditional binary occupancy grids cannot capture traversability levels, surface quality or contextual risks, leading to unsafe decisions.

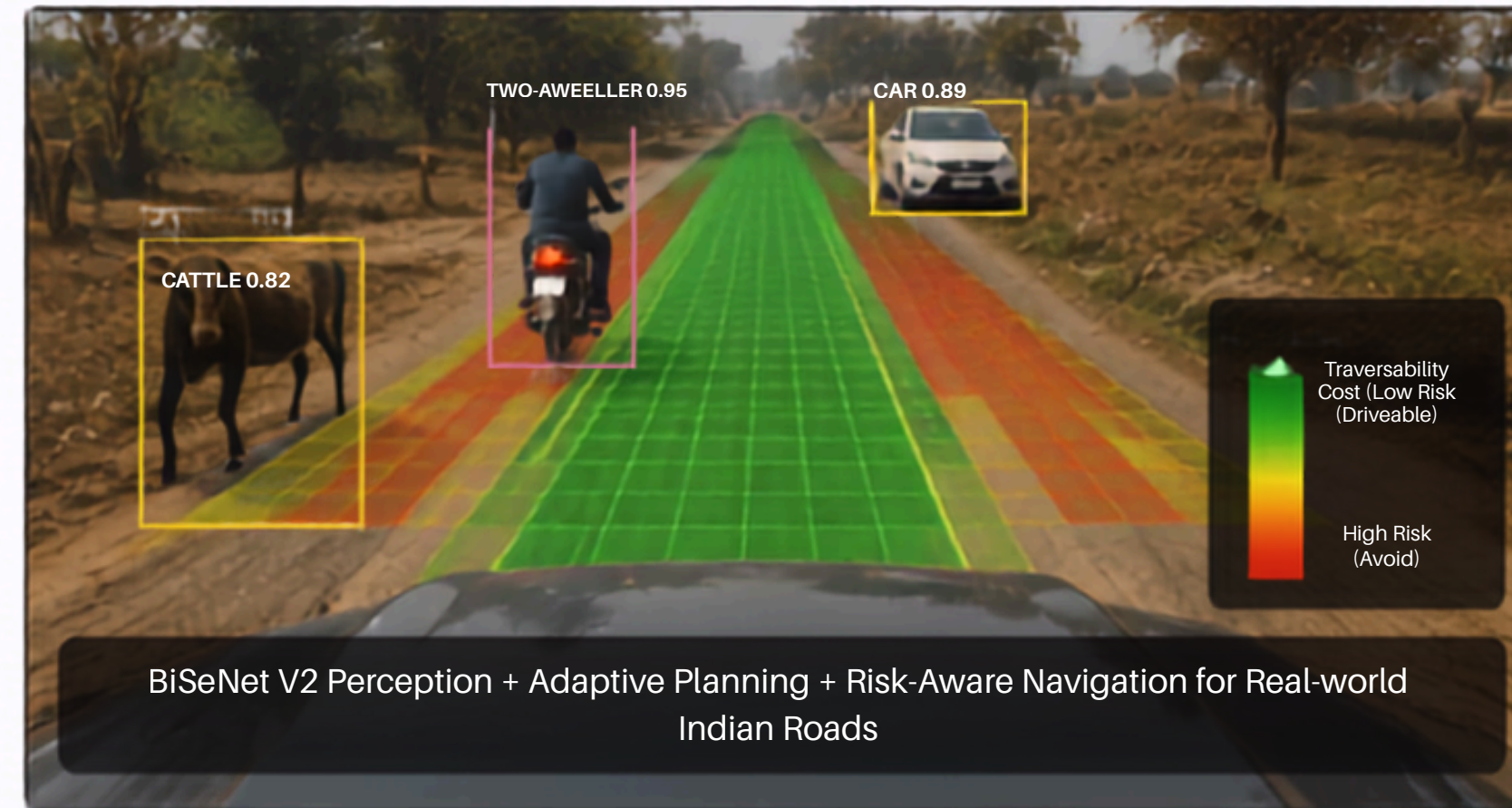


Dynamic & Unpredictable Environment: Animals, two-wheelers, pedestrians and sudden obstacles require class-aware reasoning and adaptive behavior.



Compute & Resource Constraints: Real-time, robust autonomy on edge hardware is challenging due to limited compute, memory and power.

Our Solution



Why We Stand Out

- **Model-predictive & Adaptive:** Adaptive MPPPI planner with live surface-friction estimation ensures stability on varying terrains.
- **Lane-agnostic Traversability:** Continuous traversability-risk costmap enables reliable navigation even on unstructured roads.
- **Class-aware Safety:** Asymmetric risk envelopes and RSS-based monitoring for vulnerable road users (cattle, two wheelers).
- **Production-grade Architecture:** Distributed 3-node ROS 2 stack with isolated compute for perception, planning and control.

Key Features



Lane-agnostic Traversability Costmap

Replaces binary occupancy grids with continuous risk estimation.



BiSeNet V2 Perception

Real-time semantic segmentation optimized for road, off-road & obstacles.



u-Adaptive MPPPI Planner

Adapts max. lateral acceleration using live surface friction estimation.



Class-specific Risk Envelopes

Circular buffer for cattle, teardrop for two wheelers with RSS safety monitor.



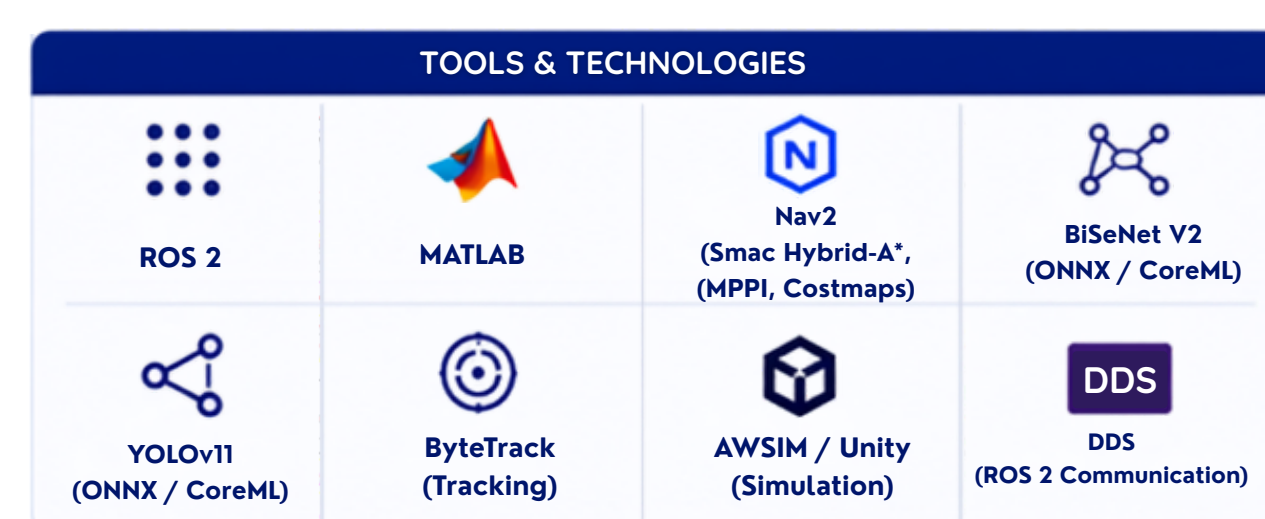
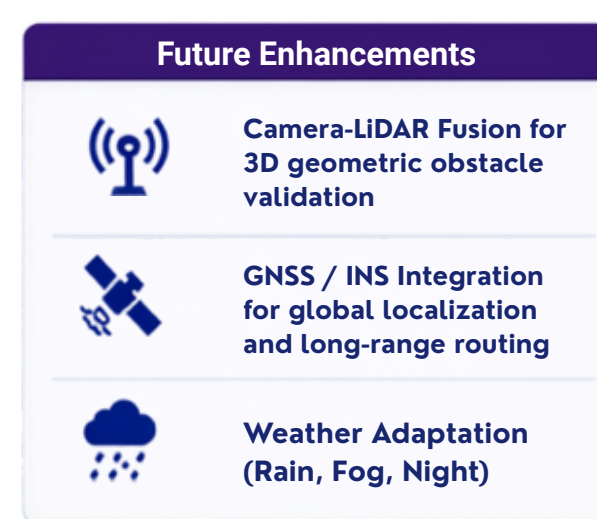
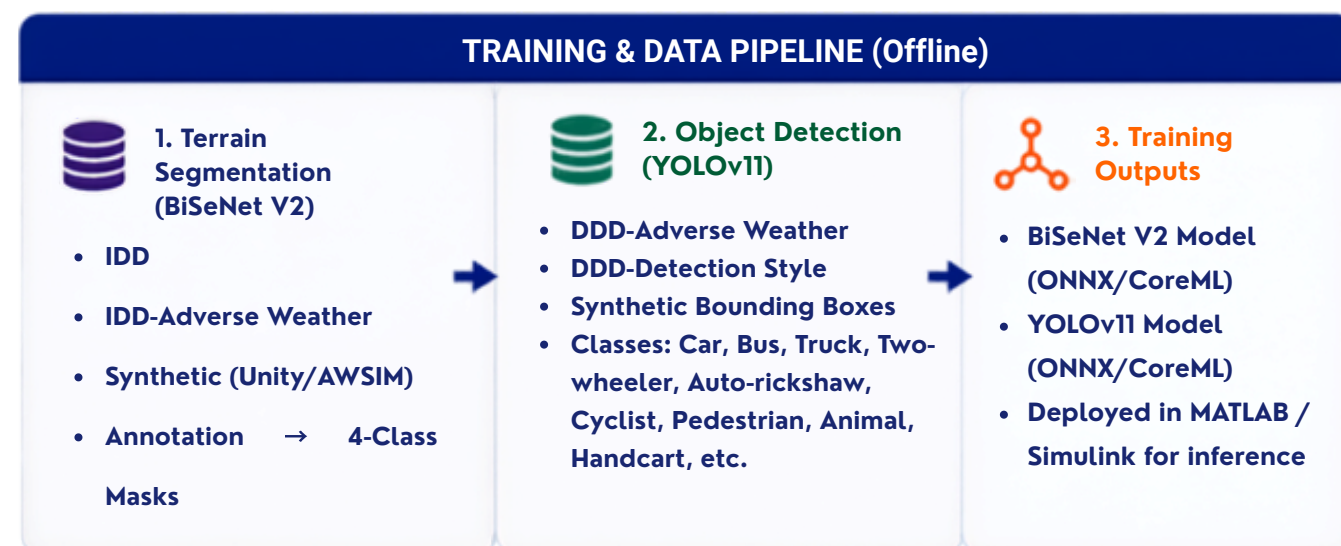
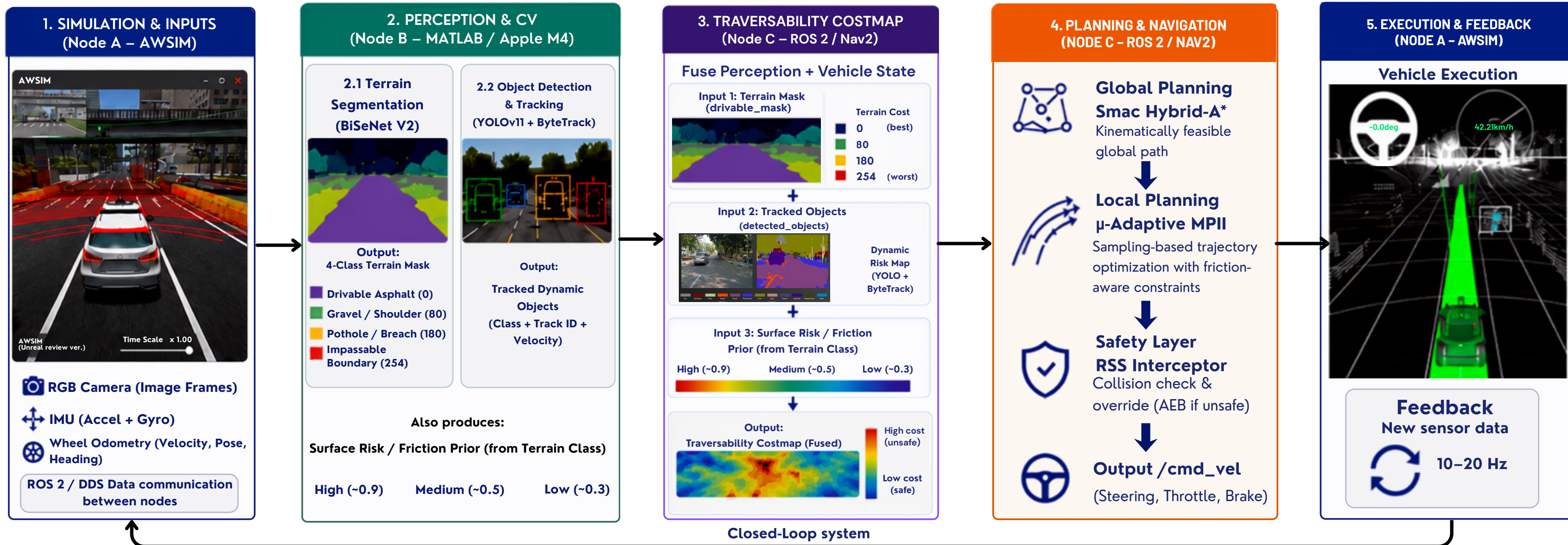
Distributed 3-node ROS 2 Stack

Removes GPU bottleneck, mirrors AV production architecture.

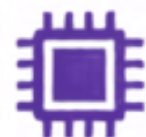


Real-time Monitoring & Alerts

Visualization dashboard with costmap, detections, trajectory & system health.



Technical Feasibility



- Proven Technology Stack: ROS 2, BiSeNet V2, YOLOv11 and MPPI for perception and planning.
- Simulation & Multi-Modal Sensing: Simulation-first validation with Camera, IMU and Wheel Odometry for robust development.
- Modular & Scalable Architecture: Real-time ROS 2 pipeline optimized for AI inference, DDS communication and future vehicle deployment.

Economical Feasibility

Parameter	Conventional Physical Approach	MARG-Nav – Simulation Approach
Validation Platform	Physical vehicle + physical road/test environment	Virtual vehicle + simulated environment
Sensor Requirement	Physical Camera / LiDAR / GNSS-IMU required	Simulated Camera / LiDAR / GNSS-IMU
Hardware Cost	~₹1.1–2.4 Lakh	vehicle/sensor are virtual
Testing & Validation Cost	Repeated field/track testing, setup and vehicle operation	Low-cost, repeatable virtual testing
Navigation Performance	Route completion, path deviation & collision rate	Route completion, collision-free rate & path efficiency
Scenario Scalability	Limited by physical vehicles, locations & test conditions	High – roads, traffic, obstacles, weather & lighting can be varied
Validation Repeatability	Repeated physical runs required	Same scenarios can be replayed consistently and

Maintenance & Safety Feasibility



- Modular & Low Maintenance: Easy updates and servicing.
- Health Monitoring: Early fault detection and diagnostics.
- Fail-Safe Safety: Collision avoidance and emergency intervention.
- Simulation & Deployment Ready: Tested virtually before vehicle integration.

Economic & Operational Viability



- Low Cost: Software-first approach minimizes hardware expenses.
- Virtual Validation: No physical vehicle needed initially.
- Scalable: Works across vehicles and road environments.
- Faster Testing: Safer validation with reduced testing costs.

Implementation Viability

Factor	Support / Approach
Manufacturability	Uses widely available components; easy to assemble and maintain.
Component Availability	100% components available in India / through local suppliers.
Serviceability	Modular design enables quick replacement and field service
Scalability	Supports add-ons like sprayers, seeders, or payload modules
Deployment	Prototype ready for field testing, pilot deployment within 3-6 months
Readiness	System tested in simulated & real-world environments, ready for scale-up

Environmental Viability



- Low Cost: Software-first development reduces hardware expenses.
- Virtual Validation: No physical vehicle needed initially.
- Scalable: Adaptable across vehicles and road conditions.
- Reduced Costs: Faster testing with lower maintenance needs.

SOCIETAL IMPACT



Safer Rural Mobility

Reduces accidents and enhances safety for farmers, pedestrians, cattle and two-wheelers.



Inclusive Access

Enables reliable mobility in remote rural regions lacking structured roads.



Livelihood Support

Improves productivity and income opportunities by enabling timely transport of people and goods.



Human & Animal Coexistence

Class-aware risk monitoring ensures safer coexistence of vehicles with animals, two-wheelers and pedestrians.

ENVIRONMENTAL IMPACT



Fuel & Energy Efficiency

Optimized planning and adaptive control reduce fuel usage and unnecessary idling.



Lower Carbon Footprint

Solar/battery hybrid power and efficient navigation minimize greenhouse gas emissions.



Sustainable Farming

Supports precision agriculture and efficient logistics, reducing resource waste (water, fertilizer, time).



Climate Resilience

Robust in dust, rain and low-light conditions for reliable year-round operations.

ECONOMIC IMPACT



Cost Reduction

Lower fuel consumption and optimized routes lead to significant operational cost savings.



Higher Productivity

Continuous, reliable operation increases field coverage and improves task completion rate.



Scalable for Agribusiness

Modular, cost-effective system suitable for small farms to large agribusiness operations.



New Market Opportunities

Enables deployment of autonomous solutions for rural mobility and smart agriculture.

BENEFITS TO STAKEHOLDERS

Farmers

Higher productivity,
lower costs

Rural Communities

Safer travel, better
connectivity

Government & Society

Supports rural innovation
and public safety

Agribusiness & Industry

Efficient operations,
scalable solutions

Environment

Lower emissions,
sustainable future

RESEARCH AND REFERENCES

RESEARCH SUMMARY

- Investigated cutting-edge semantic segmentation models to enhance perception on unstructured roads, focusing on BiSeNet V2 for real-time, low-compute performance.
- Researched adaptive motion planning using μ -Adaptive MPPI to handle varying terrain and surface conditions.
- Explored multi-sensor fusion (LiDAR, IMU, GNSS, Cameras) for reliable localization, mapping, and environmental perception.
- Analyzed RSS-based safety envelopes and open datasets/simulators for protecting vulnerable road users and enabling effective testing and validation.

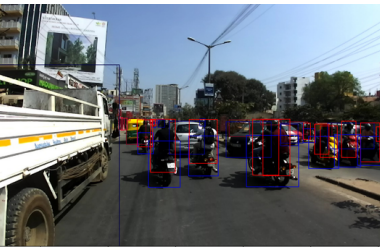
KEY TECHNOLOGIES & METHODS

- **Perception**
Bisenet V2 for real-time semantic segmentation and traversability analysis.
- **Localization**
LiDAR + IMU + GNSS sensor fusion using EKF precise 3D mapping and pose estimation.
- **Planning**
Adaptive MAPI for local path generation on varying terrain with continuous costmap.
- **Safety**
Class-specificRSS safety envelopes and decoupled safety monitor.
- **Architecture**
Distributed S-node ROS 2 stack for scalability and real-time performance.
- **Real-time Performance**
Optimized inference and planning pipeline for real-time onboard execution.

REFERENCES

- Yu, C., Gao, C., Wang, J., Yu, G., Shen, C., & Sang, N. (2021).
- Williams, G., Drews, P., Goldfain, B., et al. (2018).
- Macenski, S., Foote, T., Gerkey, B., Lalancette, C., & Woodall, W. (2022).
- Varma, G., Subramanian, A., Namboodiri, A. M., Chandraker, M., & Jawahar, C. V. (2019).

DATA SOURCES



Indian Driving Dataset (IDD):
Large-scale dataset for autonomous driving in Indian conditions.



Unity/AWSIM-based scenarios with randomized traffic, obstacles, weather & lighting for automated ground-truth generation